Towards an AI-Native Trading Strategy

hermessuen

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1 Introduction

This report introduces a framework for how to use AI agents to forecast stocks, and construct a portfolio, leading to a completely AI-Native trading strategy. We adopt an approach that is heavily influenced by how trading currently works in quantitative (mid-low) frequency hedge funds. This was a natural starting framework given the author's background.

2 A Bet

The simplest definition of an LLM is a "next-token predictor", where the "token" is a piece of text. As simple as this may seem, due to the apparent rich representational structure of language, these models are capable of "reasoning" over large amounts of text to do all sorts of useful work. Among those most relevant for the present discussion is the forecasting of financial markets. Prior to the onslaught of Big Data and Machine Learning, trading was conducted via "very smart" people ingesting lots of news and information about businesses, industries, and the economy and coming up with predictions of how stocks will trade.

LLMs should be far better than any humans at this task. Frontier AI labs have already matched or exceeded expert levels of performance on various benchmarks for forecasting, and it stands to reason that a program capable of "understanding and reasoning" over gigabytes of text has a strong chance to predict financial markets.

3 Problems

However, if one just uses LLM's directly out of the box, they tend to perform quite poorly. Here are a few of the failure modes:

- Hedging: when asked for probabilities, LLMs overly index on probabilities close to 50

- Calibration: humans are not naturally good at probability estimation, and LLMs are no different. In order to get reasonable estimates, various techniques such as Platt Scaling are required.
- Interpretability: Even with "chain of thought" reasoning, LLM's are capable of hallucinating or making illogical jumps that are not grounded in reality. Thus, they are poor at conditional probability questions. Bayesian updating of the sort: $P(B|A) = \frac{P(A|B)*P(B)}{P(A)}$ is not something they can do reliably.
- Bias: An LLM is strongly influenced by its initial prompt. If you ask it to build a Bear case for a stock it will. If you ask it to build a Bull case for a stock it also will.

Beyond getting reasonable forecasts, there is also a secondary problem of portfolio construction. It is not immediately clear that LLMs would be naturally good at this as well. Even if they were, portfolio construction is precisely an area where we should be hesitant to assume unnecessary risk. For instance we would not want an LLM hallucinate a forecast and then make an overly concentrated bet in that sector. We would like to have some safeguards and gaurantees on how our portfolio is constructed.

The most important question for an AI-Native hedge fund becomes how to best leverage LLM capabilities without falling prey to their weaknesses.

4 Towards a Framework

We have already established that LLMs are generally not proficient at pure numeric forecasting. But they are extremely good at forming summaries and analyzing text. To me, this essentially means that they are better at reasoning at a more abstract level. They are better at answering questions of the form: "What industries will experience high growth rates in the next 2 years", and "what is the market sentiment around semi-conductors stocks". Given the immense amount of context an LLM can read and consider at a given time, they should be able to provide meaningful predictions for these questions. Therefore, they should be good at understanding "the big picture" of how news, politics, and events are going to drive markets.

Likewise, LLMs should be quite good about researching and determining the precise sets of stocks that would be affected by its own predictions. For instance, if an LLM is confident that investor sentiment will continue to be optimistic towards semiconductors, it will be much better (and faster) than a human at selecting a list of assets one could use to gain exposure to that view.

Eventually we still need to make a forecast about the performance of a particular asset. A few tricks can be used to make this possible using LLMs:

- Bull and Bear: For every stock we are asked to analyze, separate agents are spawned to build both the Bull and Bear case. A supervisor agent agent combines these views
- Bounded Scores: Instead of forecasting raw returns, it is more reasonable to take scores between $[-1, 1]$. For many problems, we only need
- Confidence Score: Force the LLM to give a confidence rating for its score. This will obviously be a rough estimate, but it forces the agent to "reason" over the quality of its analysis and helps provide insight for downstream consumers to adjust the size of their bets
- Data: We cannot rely on the stored knowledge and reasoning capabilities of an LLM to produce forecasts. Those forecasts must be grounded on evidence, and come from as many independent sources as possible. Every agent will have access to many sources of real-time information and be forced to rely on what is contained there to make its predictions

With this in mind, we propose the following general framework for producing forecasts from LLMs:

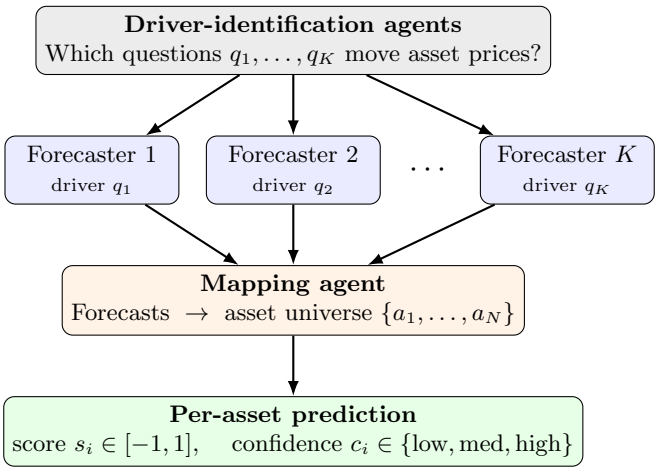


Figure 1: Workflow of the LLM forecasting system.

5 Towards a Strategy

In this section we build on the previous section to describe the formal mathematical details involved in building an AI-Native strategy.

5.1 Black-Litterman

We primarily rely on the Black-Litterman Portfolio allocation literature which describes how to robustly perform asset allocation. Modern Portfolio theory (i.e mean-variance optimization by Markowitz) essentially already answers this question. However, it is a well-known fact that the Efficient Frontier generated by a general Markowitz framework is extremely sensitive to one's forecasts over expected returns.
The Black-Litterman model was developed so that an investor does not need to input precise estimates of expected returns. Instead, an investor need only input a series of *views* over a subset of assets. These are the independent beliefs of the investor that she believes have not been reflected in the market price. Views can be absolute or relative. Below are three sample views expressed in the original paper:

View 1. International Developed Equity will have an absolute excess return of 5.25% *(confidence: 25%)*

View 2. International Bonds will outperform US Bonds by 25 basis points *(confidence: 50%)*

View 3. US Large Growth and US Small Growth will outperform US Large Value and US Small Value by 2% *(confidence: 65%)*

Note already that these views are more vague (and hence more suitable for LLMs) than statements like "Apple stock will have a return of 2 percent tomorrow".

In order to convert these views into returns, Black-Litterman starts with the Market-Implied Returns:

$$\Pi = \lambda \Sigma w_{mkt} \tag{1}$$

Where λ is the classic risk aversion parameter, Σ is the usual covariance matrix, and w_{mkt} are the market-cap weights for each asset. Black Litterman then makes the following assumptions:

$$\mu \sim \mathcal{N}(\Pi, \tau\Sigma), \qquad Q = P\mu + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \Omega) \tag{2}$$

P matrix identifying the assets involved in the views ($K \times N$; $1 \times N$ for a single view)
Ω diagonal covariance matrix of view error terms, the uncertainty in each view ($K \times K$)
Π Implied Equilibrium Return Vector ($N \times 1$)
Q View Vector ($K \times 1$)
τ scalar scaling the uncertainty of the prior: $\tau\Sigma$ is the covariance of the estimate of the mean, so small τ (typically 0.01–0.05) means high confidence in the equilibrium returns Π

In words, this is essentially the view that expected returns follow a normal distribution centered around the market implied returns and that our view vector Q is equivalent to selecting some number of the assets in our universe, weighting them, multiplying by their mean, subject to noise coming from the uncertainty of our view. τ tells us how uncertain we are of the prior. In the most extreme case, we are overly confident in our own view (i.e the omega matrix is an identity) and the optimal asset allocation should shift wildly from the implied equilibrium

From there, Black-Litterman tries to maximize the Posterior Distribution density:

$$p(\mu \mid Q) \propto \exp\left[-\tfrac{1}{2}\left((\mu-\Pi)^\top(\tau\Sigma)^{-1}(\mu-\Pi) + (Q-P\mu)^\top\Omega^{-1}(Q-P\mu)\right)\right] \tag{3}$$

Taking the derivative and solving the first order condition, we get the formula for the Black-Litterman return vector:

$$\mu_{BL} = \left[(\tau\Sigma)^{-1} + P^\top\Omega^{-1}P\right]^{-1}\left[(\tau\Sigma)^{-1}\Pi + P^\top\Omega^{-1}Q\right] \tag{4}$$

5.2 LLM Adaptation

The Black-Litterman formulation above provides a straightforward solution to our LLM forecasting problem. We can convert LLM z scores (which are from $[-1, 1]$), into some estimate for alpha:

$$\alpha_i = \text{IC} \cdot \sigma_i \cdot z_i \tag{5}$$

Here, IC is the Information Coefficient of our LLM. This can be fit on historical data. This essentially represents our "view". There are many different methods for specifying the matrix P that "select" which stocks to be multiplied by our view. For the 3 sample views given above by Black-Litterman, here is an example P matrix:

$$P = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ -1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.9 & -0.9 & 0.1 & -0.1 & 0 & 0 \end{bmatrix} \tag{6}$$

You can see that in the case of relative views, the weights sum to 0 while in the case of an absolute view, the weights sum to 1. In our case to start, we stick to absolute views given by the α_i, although future research on relative views must be conducted.
Finally, we need to specify the Ω. This is done by setting each view to the variance of the portfolio set by that view:

$$\Omega = \tau \cdot \text{diag}\big(P\Sigma P^\top\big) \tag{7}$$

Note that by using this choice, τ ultimately drops from the equation. This gives us all of the variables and information required to formulate an expected return vector from an LLM forecasting system that is robust interpertable. It essentially answers:

> *How do I optimally create a return vector given the current market implied return and my own beliefs, subject to my own confidence?*

5.3 Portfolio construction

Most commonly for portfolio construction, one should adopt a mean-variance trade off. Formally, one seeks to solve the following optimization problem:

$$\max_w \; w^\top \mu - \frac{\lambda}{2} w^\top \Sigma w \tag{8}$$

For an unconstrained problem it has an easy closed-form solution:

$$w^* = \frac{1}{\lambda} \Sigma^{-1} \mu \tag{9}$$

Here, u is constructed from Black-Litterman.

However, for our case we would like to be a long-only fund (to avoid risk) and have our weights sum to 1. We would like to reach a target book and allocate our capital accordingly. We may also have constraints on the amount of risk associated with any particular asset. Therefore our problem may look something like this:

$$\begin{aligned} \max_w \quad & w^\top \mu_{BL} - \frac{\lambda}{2} w^\top \Sigma w & \\ \text{s.t.} \quad & \mathbf{1}^\top w = 1 & \text{(full investment)} \\ & w_i \geq 0 \quad \forall i & \text{(long only)} \\ & \sqrt{w^\top \Sigma w} \leq \sigma_{\max} & \text{(risk budget)} \end{aligned} \tag{10}$$

This can be solved in a straightforward way since it is a convex problem (since we do not have any market impact). There are a few details for our initial solution. First, the Covariance matrix Σ is notoriously singular. Therefore, it is much better to decompose it in terms of Factor loadings:

$$\Sigma = B\Sigma_f B^\top + D \tag{11}$$

Second, a major problem for us is the universe. Suppose our universe is the Russell 2000. It is simply infeasible to come up with LLM forecasts for every single one of those assets. Recall that in our original framework, we had a step that outlined the sets of stocks for which the LLM theses is tied to. In order to take advantage of this view, we essentially only perform the mean-variance optimization with respect to this restricted set of assets. Since this can have the unfortunate side-effect of liquidating all of our previous positions, we can also augment the universe to include our current holdings and force an LLM to make predictions on the the larger universe set.

6 Open Questions and Research Directions

These are just the preliminary sketches of how one could orchestrate an AI-native trading strategy. However, there are many loose ends. Below are some areas of investigation that I believe are worth immediately pursuing:

- Relative forecasts: Black-Litterman is arguable better posed for using relative views as opposed to absolute ones. LLMs may also be better at proposing relative forecasts
- Data: Input information quality remains the highest lever one can pull to improve the performance of any forecasting system.
- Universe Construction: There needs to be a faster way to generate a more usable universe for trading. It is infeasible to produce forecasts for thousands of stocks via an LLM - however it is also not desirable to only trade a section of your universe at any given point. One can imagine a statistical or RL approach that balances the cost of research vs the potential gains from its forecasts
- Fine-Tuning: We could increase an LLMs performance for forecasting by fine-tuning its last layer to produce a numeric output